PART 0 · SHEET COVER
COVER

AI-PLAYS-POKE

AN AGENT THAT PLAYS POKEMON BLUE BY UNDERSTANDING

TWO PERCEPTION CHANNELS - ONE GRID - DIFFED EVERY FRAME

This handbook describes the system as built and measured: the emulator bridge, the RAM reader, the vision channel, the decision core, the operator console, the fleet that builds and verifies it, and what watches it. Every number was taken from the code or measured on the live system; the ledger behind it is specs/_facts.md.

The set states its own defects where it measured them: the one-row grid offset (R-001), the idle-spin admission waiver, the people-identification failure rate. A page that hides a measured defect is marketing, not a handbook.

Document	AP1A-03, Issue 1
Scope	the agent, its perception, its console, its fleet
The one number	72.2 pc cell agreement, best measured
Readers	worker, judge, operator
Basic date	24 September 2026
Change date	11 November 2026
Sheets	9 plus this cover

AIP

PART 1 · SHEET 1.1-1

SYSTEM OVERVIEW

1.1.1 INTRODUCTION.

AI-PLAYS-POKE is an agent that plays POKEMON BLUE by understanding each room. It perceives through two channels, RAM and a vision model, that must agree cell for cell in one grid and one legend; JEV decides most cycles and hands the state to a reasoning model only when it cannot decide (figure 1.1-1).

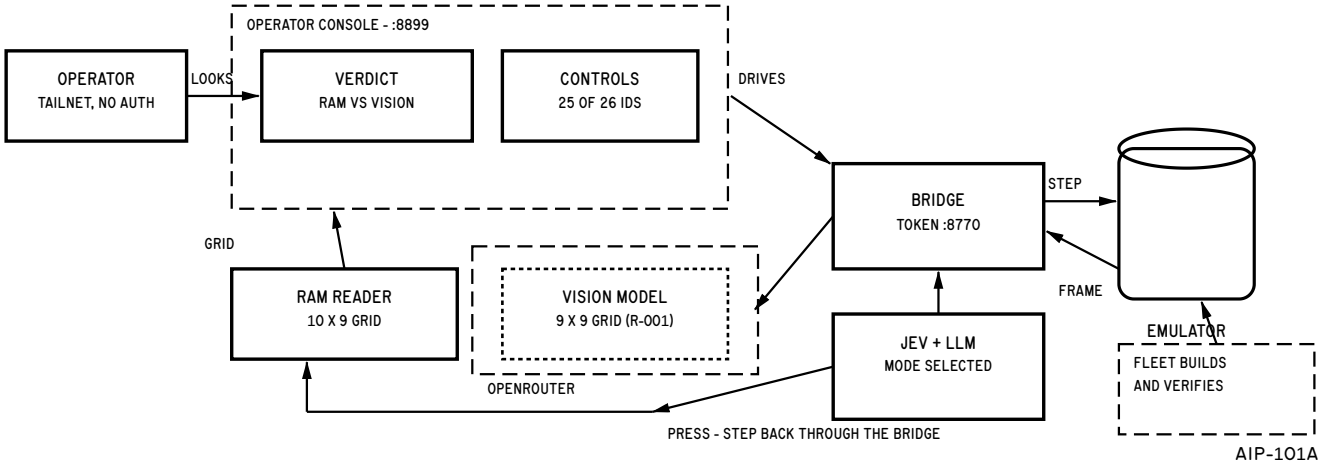


Figure 1.1-1. The System As Built, With The Vision Grid Defect Dotted

1.1.2 FUNCTIONAL DESCRIPTION.

The emulator advances only when the bridge delivers inputs, so a live ticker stands in for a paused hand-held. The operator watches and drives through the console; the agent runs through the same bridge, and the two never drive at once. The fleet builds, tests and releases the agent; the board and the judge keep its work honest.

1.1.2.1 State Of The Goal.

The September run reached VIRIDIAN CITY in 28 episodes. The October 27 run stopped the morning it started: 4 episodes, the last inside RED'S HOUSE 1F. The machine has improved; the game has not advanced since (measured 2026-11-11).

1.1.2.2 The Fleet Around It.

Ten lanes serve this project: a foreman at priority 10 on a six-hour pin, plus qa, sync, releng, docs, readme, review, perf and their satellites. Since September 28 the foreman has landed 60 commits; the board holds 357 rows, 339 complete. The lanes share this host and its free-model quota with the emulator.

1.1.2.3 What Is Deliberately Absent.

No story modelling, no scripted walk tables, no voice control. The agent must win by understanding; everything else is scaffolding and is described only as it serves that.

1.1.3 PERFORMANCE AND DESIGN DATA.

Table 1.1-1. Where The Goal Stands

Parameter	Value
Goal, set by operator	way past PALLET TOWN in 2 months
Sept run (28 eps)	VIRIDIAN CITY reached
Oct 27 run	4 eps, RED'S HOUSE 1F, then stopped
Last game action	2026-10-27 08:49
Console position	PALLET TOWN 5,6 down
Version	v1.1.0 published 2026-11-04

WARNING

THE CONSOLE HAS NO AUTHENTICATION. ANYTHING ON THE TAILNET CAN DRIVE THE EMULATOR AND SPEND MODEL QUOTA.

```
START A RUN                                  OPERATOR
1 BRIDGE - verify health op
2 LIVE - set OFF
3 MODE - pick JEV/LLM/HYBRID
  √ LOG - mode echoed at run head
5 RUN - cron_runner --cycles 20
  √ VERDICT - agree/conflict shown
```

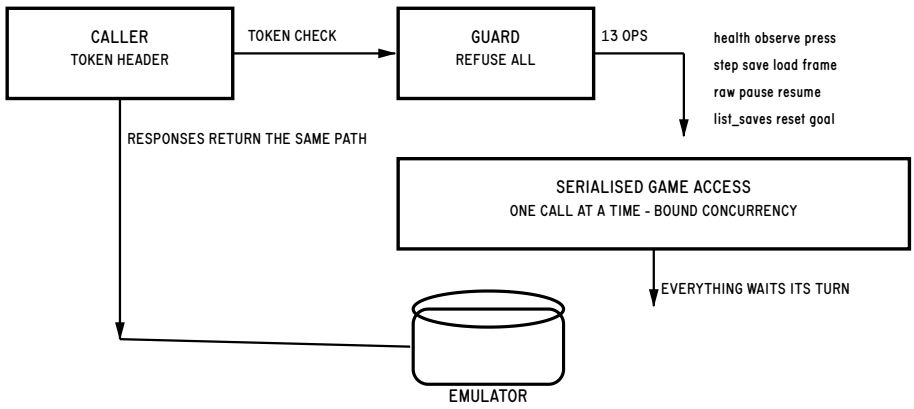
OVR

PART 2 · SHEET 2.1-1

EMULATOR BRIDGE

2.1.1 INTRODUCTION.

The bridge is the only path to the emulator. It binds loopback only, refuses every request that does not carry its token, and exposes thirteen operations. A session without the token cannot talk to the emulator at all (figure 2.1-1).



AIP-201A

Figure 2.1-1. Token Gate, Operation Dispatch, Serialised Game Access

2.1.2 FUNCTIONAL DESCRIPTION.

Every caller presents the token; the guard compares it in constant time and refuses the rest. Operations are line-delimited requests on one socket or single commands. Game access is serialised: one call at a time, so a console viewer and an agent cycle queue rather than corrupt the emulator.

2.1.2.1 Serialisation.

An unbounded thread per connection once drove the host to load 1016. The bound now caps concurrency and Game access is a lock, so one slow frame makes everything wait its turn rather than fight.

2.1.2.2 One Socket Or One Command.

A live session speaks line-delimited requests on one socket; everything else sends single commands. When one Game call stopped answering it held the bridge for 25 seconds and every panel starved; the bound now makes one slow frame cost itself only.

2.1.2.3 Revocation.

The bridge refuses to boot without a token file, so it cannot start naked by accident. Revoke by killing the process and deleting the token; enable per session.

2.1.3 PERFORMANCE AND DESIGN DATA.

Table 2.1-1. Bridge Data

Parameter	Value
Listen	127.0.0.1:8770, loopback only
Auth	token, compare_digest
Operations	13 (health ... goal)
Frame, native	160 x 144 px, about 2.7 kB
Frame when degraded	412 B blank - restart bridge
Step speed, measured	120 frames in 6 ms
Capacity used at 60 fps	about 0.3 percent

CAUTION

TWO DRIVERS ON ONE BRIDGE FIGHT. STOP THE CONSOLE LIVE TICKER BEFORE AN AGENT RUN.

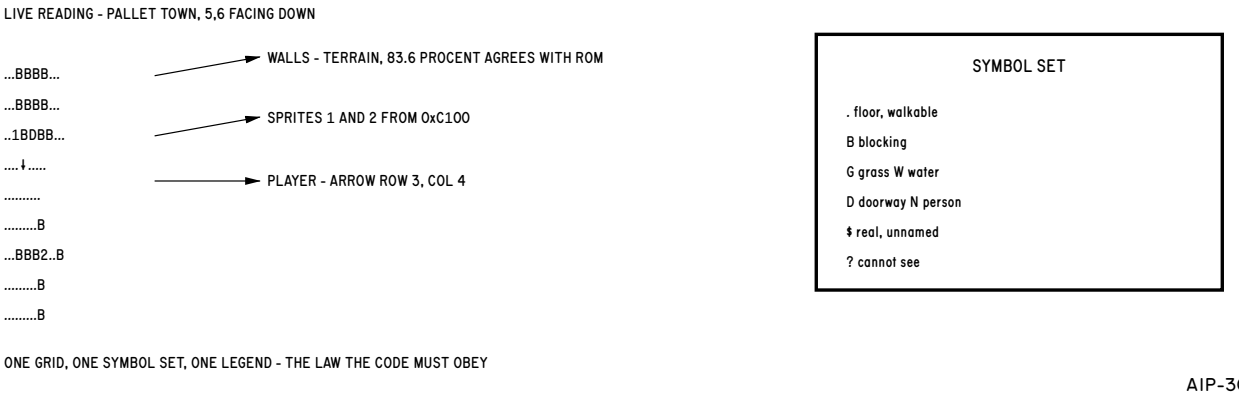
BRG

PART 3 · SHEET 3.1-1

RAM READER

3.1.1 INTRODUCTION.

The RAM reader is System One's eye. It reads the emulator memory directly: map, player tile, facing, party, collision, and the sprite state table. It renders the room as a 10 by 9 tile grid with the player's facing arrow in his own cell (figure 3.1-1).



AIP-301A

Figure 3.1-1. The Live Tile Grid And Its Legend, With Measured Cells

3.1.2 FUNCTIONAL DESCRIPTION.

Terrain comes from the ROM tile table; sprites come from wSpriteStateData1 at address C100, sixteen records of sixteen bytes, the player in slot zero. Facing renders as an arrow. The grid is the reference the vision channel is judged against, and its own terrain labels measured 83.6 percent against ROM collision.

3.1.2.1 Reference, Not Truth.

The reader is a reference, not an oracle. Sprite meaning, dialogue and menus are invisible to it. A verdict disagreement is a finding to investigate, never a reason to overwrite one channel with the other.

3.1.2.2 What The Reader Cannot See.

Text, menus, dialogue and name entry are invisible to RAM. A sprite's meaning is not in memory either. An out-of-range cell answers walkable, so read the code before trusting a verdict row that says the way is clear.

3.1.2.3 Cost.

The read is free and runs every cycle, which is exactly why it is the reference channel and the vision model is the guest that must justify its 25,000 tokens.

3.1.3 PERFORMANCE AND DESIGN DATA.

Table 3.1-1. Reader Data

Parameter	Value
Grid	10 cols x 9 rows, arrow in player cell
Cell	16 px, origin measured on screen
Sprite table	0xC100, 16 x 16 bytes
Player slot	0, excluded from people probes
Terrain vs ROM	83.6 percent agreement
Cost per read	free, every cycle

```
TAKE A READING                    OPERATOR
1 BRIDGE - verify health
2 STATE - read /api/state
  √ GRID - 9 rows of 10
4 ARROW - confirm facing
```

RAM

PART 4 · SHEET 4.1-1

VISION CHANNEL

4.1.1 INTRODUCTION.

The vision channel describes the screen the way the RAM reader describes the room: one character per cell. It is the only channel that can name what a thing is - a person, a doorway, grass - and the only channel that hallucinates (figure 4.1-1).

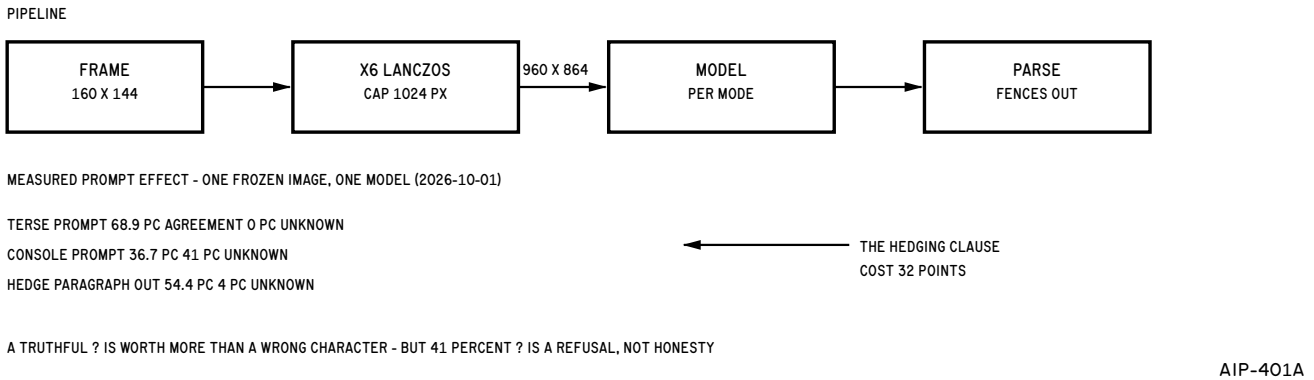


Figure 4.1-1. Pipeline And The Measured Effect Of Prompt Wording

4.1.2 FUNCTIONAL DESCRIPTION.

The frame is enlarged six times with LANCZOS smoothing, capped at 1024 pixels, and sent to the configured model with a terse grid prompt. The reply is parsed through code fences, clamped to nine rows of ten characters, and compared against the RAM grid. Model, provider and prompt are measured variables, not tastes.

4.1.2.1 Models Are Multi-Possible.

AIPP_VISION_MODELS carries an ordered list; the default served model is GPT-4O-MINI at 2.4 seconds and 25k tokens. KIMI K3, 2.8 trillion parameters, measured 62.2 percent at 97 seconds - bigger and newer was worse.

4.1.2.2 People Are A Different Task.

The people probe scored exact sprite cells against wSpriteStateData1: one frame in three correct, and the model answered girl at cell 1,4 on all three frames including the two where no girl stands there. It anchors. Constrain the question to RAM-predicted cells or do not trust the answer.

4.1.2.3 Cost Discipline.

A reading costs about 25,000 input tokens. The auto trigger reads on movement or scene change only; the 120-second idle timer was removed on 2026-10-01 after it generated readings around the clock for nothing.

4.1.3 PERFORMANCE AND DESIGN DATA.

Table 4.1-1. Vision Data

Parameter	Value
Default model	openai/gpt-4o-mini
Reading time	2.4-4 s
Tokens per read	about 25k in, 34-100 out
Best cell agreement	72.2 percent, terse prompt
Structure agreement	scored on non-floor cells
People: frames correct	1 of 3 (fails 66.7 pc)
People: anchoring	same wrong answer on 2 of 3

CAUTION

A READING THAT CANNOT ANSWER USES ? - BUT 41 PERCENT ? IS A REFUSAL. RETUNE THE PROMPT, NOT THE MODEL.

VIS

PART 5 · SHEET 5.1-1

DECISION CORE

5.1.1 INTRODUCTION.

JEV is System One: it answers every cycle and says when it cannot. The LLM is System Two: it repairs the agent's state on escalation and never presses a button. The two are modes of one core, selected explicitly, never inherited silently (figure 5.1-1).

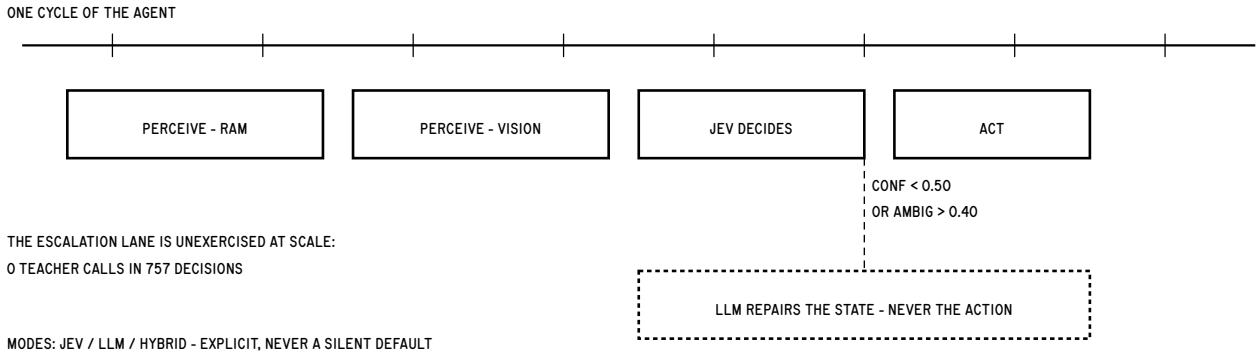


Figure 5.1-1. One Cycle, With The Escalation Lane Dotted

5.1.2 FUNCTIONAL DESCRIPTION.

The mode resolves from AIPP_DECISION_MODE, then CRON_DECISION_MODE, by exact spelling; legacy spellings stay byte-identical in the episode log so old runs stay comparable. Thresholds are operator-fixed: hand off below 0.50 confidence or above 0.40 ambiguity.

5.1.2.1 The Default That Cannot Walk.

The shipped default, JEV alone, left PALLET TOWN in 0 of 5 tries; the pure LLM did 6 of 6. A run launched without an explicit mode must refuse, not degrade to the arm that cannot leave town.

5.1.2.2 Evidence From The September Run.

757 decisions across 28 episodes, 6 escalations. Autonomy 0.99 counts decisions not escalated; participation 0.99 counts cycles answered. They agree here only because the teacher was never called. They are different numbers and are reported separately.

5.1.2.3 What JEV Cannot Decide.

Map topology. All six escalations clustered on the Route 1 to Viridian transition, and episode 28 crossed it unaided. The next run must cross a boundary it has never seen, or the number proves nothing new.

5.1.3 PERFORMANCE AND DESIGN DATA.

Table 5.1-1. Decision Data

Parameter	Value
Modes	jev / llm / hybrid, explicit
Env	AIPP > CRON > refuse
Handoff confidence	0.50
Handoff ambiguity	0.40
Sept run	757 decisions, 6 escalations
Teacher calls	0 in 757 - path unexercised
Legacy spelling	byte-identical (M6)

CAUTION

A SILENT DEFAULT ARM IS A RUN THAT WALKS INTO A WALL ALL DAY. REFUSE WITHOUT A MODE.

DEC

PART 6 · SHEET 6.1-1

OPERATOR CONSOLE

6.1.1 INTRODUCTION.

The console is the operator's eyes and hands: thirteen panels, the live game, the two grids, their verdict, and every control with a name derived from what it does. It drives the same bridge the agent drives (figure 6.1-1).

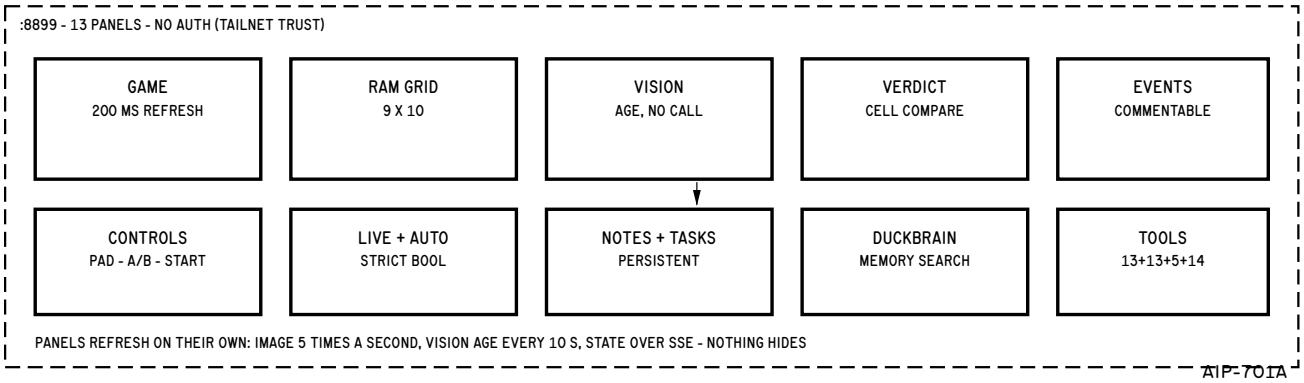


Figure 6.1-1. Panel Layout And The Refresh Law

6.1.2 FUNCTIONAL DESCRIPTION.

The game image refetches five times a second while the world runs; the vision panel ticks its age every ten seconds without spending a model call; state arrives over server-sent events. A failure answers a 500 body instead of dropping the socket, so no panel ever blanks.

6.1.2.1 Honest Input.

Toggles read real booleans only: TRUE/FALSE/1/0/YES/NO/ON/OFF. Anything else leaves the state unchanged - a malformed request once switched the world on and blanked the video minutes later.

6.1.2.2 Naming Is Load-Bearing.

Seventeen controls carried no id until 2026-10-01, which is why the mobile tap-size defect stood for weeks: nothing could select them. The duplicate-id check must read the whole page, not only the controls - a button and a panel once both claimed vision.

6.1.2.3 Chat.

The chat box posts into the agent's own session. One probe used the wrong field name, read an empty-error reply, and nearly reported the wire broken. Verify the field before blaming the wire.

6.1.3 PERFORMANCE AND DESIGN DATA.

Table 6.1-1. Console Data

Parameter	Value
Listen	:8899, CONSOLE_PORT
Panels	13
Controls	26, 25 with stable ids
Image refresh	200 ms while live
Vision age tick	10 s, no model call
Tool surface	13 bridge + 13 api + 5 mem + 14 cli
Auth	none - tailnet only

WARNING

THE CONSOLE IS UNAUTHENTICATED AND CAN DRIVE THE EMULATOR AND SPEND QUOTA. EXPOSE IT ON THE TAILNET ONLY.

RECOVER A BLANK PICTURE **OPERATOR**

```
1 SYMPTOM - frame 412 bytes
2 LIVE - set OFF
3 BRIDGE - restart service
  √ FRAME - about 2.7 kB again
5 VISION - ask now
```

CON

PART 7 · SHEET 7.1-1
FLEET BUILD-AND-VERIFY

7.1.1 INTRODUCTION.

The fleet builds and verifies the agent: a scheduler admits a foreman by board work, the foreman dispatches one worker per tick, a judge grades the evidence, and the board remembers. Nothing lands on a claim (figure 7.1-1).

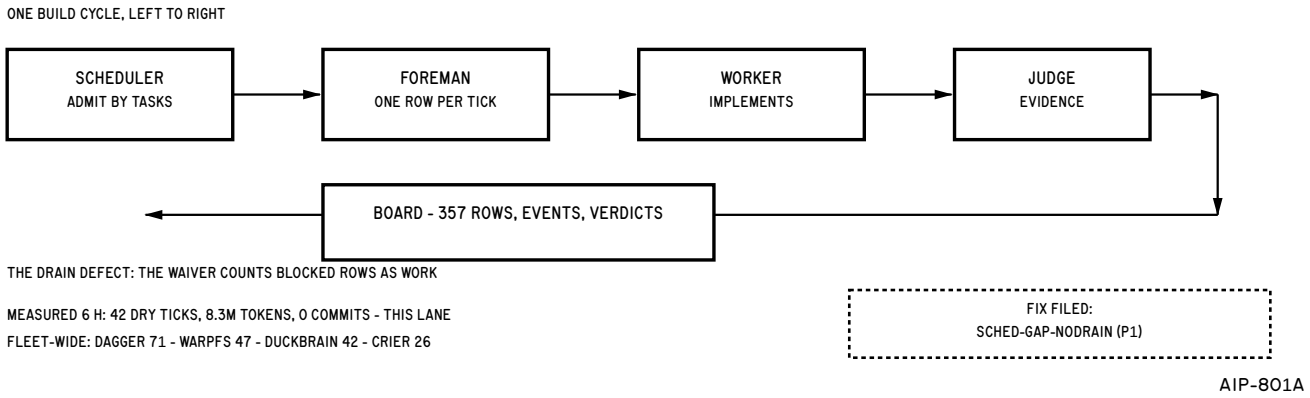


Figure 7.1-1. Scheduler To Judge To Board, With The Idle-Spin Defect Dotted

7.1.2 FUNCTIONAL DESCRIPTION.

The lane's admission mode is tasks: a non-empty board waives the six-hour cooldown pin. Workers run in worktrees with their own environments; guards run at commit; releases are cut only on owner authorisation. Verification receipts and verdicts land beside the code they grade.

7.1.2.1 Idle Spin, Measured.

Drained must mean no WORKABLE row. It currently means no open row: blocked and dep-gated rows waived the cooldown and produced 42 dry ticks in six hours on this lane alone - 8.3 million input tokens for nothing.

7.1.2.2 What A Good Tick Is.

A tick that reports nothing changed, honestly, in one line, is a good tick. A tick that re-audits finished work is a wasted one. The foreman prompt says so; the admission loop must not fight it by re-admitting every three minutes.

7.1.2.3 The Judge.

Tier 1 runs secrets, static analysis and tests; tier 2 grades the evidence. Judge runs have been refused with a 402 billing fault, and a gate whose judge never sat is recorded DEGRADED, never PASS.

7.1.3 PERFORMANCE AND DESIGN DATA.

Table 7.1-1. Fleet Data

Parameter	Value
Board	357 rows, JSONL, git-backed
This project, complete	339 of 357
Blocked	2 (owner-gated, dep-gated)
Pending, real work	16 incl. ARM-2, VIS-MODEL-1
Dry ticks, 6 h	42 (8.3M tokens)
Commits since 09-28	60 by 2026-10-01

CAUTION

A BOARD THAT LOOKS NON-EMPTY KEEPS A TASKS LANE SPINNING. MARK BLOCKED ROWS BLOCKED.

FLT

PART 8 · SHEET 8.1-1
MONITORING AND RELEASE

8.1.1 INTRODUCTION.

The agent is observed from four directions: Prometheus metrics on the dashboard, a node_exporter record from the cron lane, narration keys in DuckBrain, and the scheduler's own tick and cost ledger (figure 8.1-1).

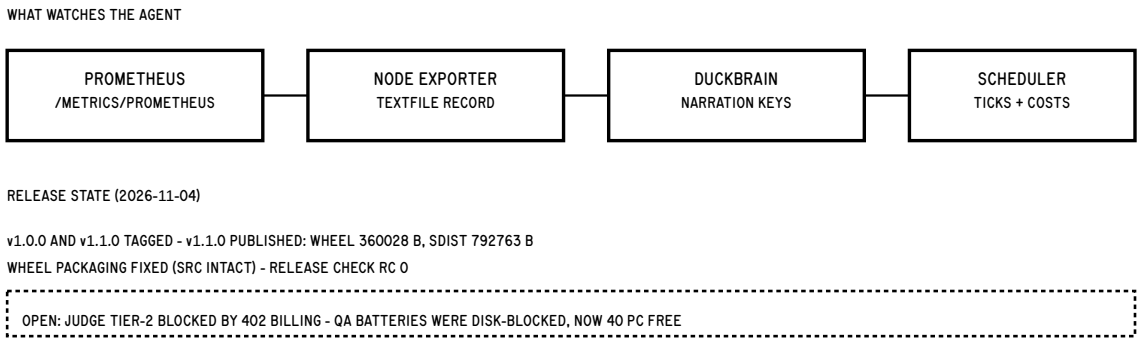


Figure 8.1-1. Observability Surfaces And The Release State

8.1.2 FUNCTIONAL DESCRIPTION.

Version 1.1.0 is the first published release: the tag fires its own CI, the wheel installs with the source package intact, and the version surfaces agree. Cuts require owner authorisation; the releng lane audits readiness each sweep.

8.1.2.1 Cost Truth.

Costs are counted from the scheduler's session ledger by base URL and model, never from a billing label that a gateway restart can restamp.

8.1.2.2 What The Numbers Must Never Do.

A cost read from a billing label lies after a gateway restart; count by base URL and session ledger instead. A lazy all-floor answer must never win a cell-match metric: score the cells that carry meaning, the walls and doors and sprites.

8.1.3 PERFORMANCE AND DESIGN DATA.

Table 8.1-1. Release Data

Parameter	Value
Version	1.1.0, pyproject + CHANGELOG agree
Wheel	360,028 bytes, src intact
sdist	792,763 bytes
Tag CI	success on the tag commit
Judge tier 2	blocked, 402 billing
QA batteries	disk was 98 pc, now 40 pc

CAUTION

AN INSTALLED WHEEL FROM v1.0.0 OR v1.1.0 BEFORE THE FIX IS NOT USABLE. REBUILD AFTER 70c272b.

MON

PART 9 · SHEET 9.1-1
FLIGHT DATA FILE

9.1.1 INTRODUCTION.

This part collects the operator procedures. Each control is named as placarded on the console; each verification line starts with a check. Nothing here requires editing a file.

START A RUN	OPERATOR
1 BRIDGE - verify health op	
2 LIVE - set OFF	
3 MODE - pick JEV/LLM/HYBRID	
√ LOG - mode echoed at run head	
5 RUN - cron_runner --cycles 20	
6 VISION - ask now (25k tok)	
√ VERDICT - agree/conflict	

RECOVER VIDEO	OPERATOR
1 SYMPTOM - frame 412 bytes	
2 LIVE - set OFF	
3 BRIDGE - restart service	
√ FRAME - about 2.7 kB again	
√ STATE - tile unchanged	

TAKE A READING	OPERATOR
1 VISION - ask now	
√ VMETA - tokens + finish	
√ GRID - 9 rows of 10	
4 VERDICT - read conflicts	
5 NOTE - record the finding	

END A SESSION	OPERATOR
1 RUN - let the cycle end	
2 LIVE - set OFF if left on	
3 SAVE - bridge save slot	
√ SAVES - list shows the slot	
5 NOTE - what you saw	

9.1.2 RULES THAT GOVERN EVERY PROCEDURE.

Never drive the emulator from two places at once. Never let a run start without an explicit mode. A finding is a board row, not a memory.

FDF

FILE A FINDING	OPERATOR
1 FINDING - name the row	
2 EVIDENCE - command + output	
3 BOARD - append, never chat	
4 √ VALIDATE - boardctl ok	

RELEASE A CUT	OPERATOR
1 AUTH - owner says cut	
2 CHECK - release_check rc 0	
3 TAG - annotated, on main	
4 √ CI - green on the tag	
5 √ ASSETS - wheel installs	